

Check Tic Tac Toe 🐍🐍

Exercise 26 (and [Solution](#))

This exercise is Part 2 of 4 of the Tic Tac Toe exercise series. The other exercises are: [Part 1](#), [Part 3](#), and [Part 4](#).

As you may have guessed, we are trying to build up to a full tic-tac-toe board. However, this is significantly more than half an hour of coding, so we're doing it in pieces.

Today, we will simply focus on checking whether someone has WON a game of Tic Tac Toe, not worrying about how the moves were made.

If a game of Tic Tac Toe is represented as a list of lists, like so:

```
game = [[1, 2, 0],
        [2, 1, 0],
        [2, 1, 1]]
```

where a 0 means an empty square, a 1 means that player 1 put their token in that space, and a 2 means that player 2 put their token in that space.

Your task this week: given a 3 by 3 list of lists that represents a Tic Tac Toe game board, tell me whether anyone has won, and tell me which player won, if any. A Tic Tac Toe win is 3 in a row - either in a row, a column, or a diagonal. Don't worry about the case where TWO people have won - assume that in every board there will only be one winner.

Here are some more examples to work with:

```
winner_is_2 = [[2, 2, 0],
               [2, 1, 0],
               [2, 1, 1]]
```

```
winner_is_1 = [[1, 2, 0],
               [2, 1, 0],
               [2, 1, 1]]
```

```
winner_is_also_1 = [[0, 1, 0],
                    [2, 1, 0],
                    [2, 1, 1]]
```

```
no_winner = [[1, 2, 0],
              [2, 1, 0],
              [2, 1, 2]]
```

```
also_no_winner = [[1, 2, 0],
                  [2, 1, 0],
                  [2, 1, 0]]
```